

EXERCISE 2 SQL AGGREGATES & OPERATORS

Section 1

1.

department
IT
HR
FINANCE

```
SELECT DISTINCT department
FROM students;
```

2.

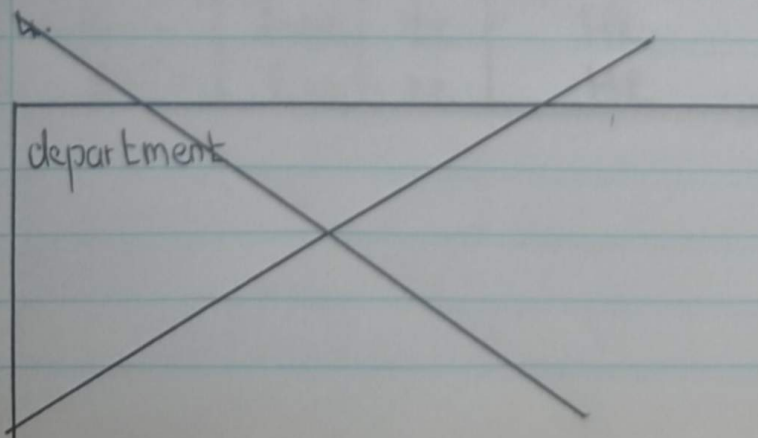
department	avg-age
IT	
HR	22
FINANCE	23

```
SELECT department,
Avg-age
FROM students
GROUP BY department;
```

3.

department	Student-Count
IT	2
HR	2

```
SELECT department,
Student-Count
FROM Students
GROUP BY department
HAVING COUNT(*) > 1;
```



4.

```
SELECT Student-id,
        Name,
        age,
        department
FROM Students
WHERE age BETWEEN 21 AND 23;
```

Student-id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	FINANCE
5	Eve	22	HR

5.

```
SELECT Student-id,
        Name,
        age,
        department
FROM Students
WHERE department IN ('IT', 'HR') AND age > 21;
```

Student-id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

SECTION 2

6.

```
SELECT department,  
        total_Credits  
FROM Courses  
GROUP BY department  
HAVING SUM(Credits) > 5;
```

department	total_Credits
IT	11

7.

```
SELECT Course-id,  
        Course-name,  
        department,  
        Credits  
FROM Courses  
WHERE Credits != 4;
```

Course-id	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	FINANCE	2
105	Statistics	HR	3

4.

```
SELECT Student-id,
       Name,
       age,
       department
FROM Students
WHERE age BETWEEN 21 AND 23;
```

Student-id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	FINANCE
5	Eve	22	HR

5.

```
SELECT Student-id,
       Name,
       age,
       department
FROM Students
WHERE department IN ('IT', 'HR') AND age > 21;
```

Student-id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

8.

```
SELECT Course-id,
       Course-name,
       Credits
FROM Courses
ORDER BY Credits DESC
LIMIT 3;
```

Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

SECTION 3

9.

```
SELECT max-grade,
       min-grade,
       avg-grade
FROM enrollments
```

max-grade	min-grade	avg-grade
90	78	84.85

10.

```
SELECT Course-id,
        enrollment-count
FROM    enrollments
GROUP BY Course-id;
```

Course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

Course-id	enrollment-count	enrollment-count
101	1	1
102	1	1
103	1	1
104	1	1
105	1	1

SECTION 4

11.

```
SELECT department,
        total-salary,
        total-bonus
FROM    Salaries
GROUP BY department;
```

department	total-salary	total-bonus
IT	122 000	10 500
HR	109 000	7 500
FINANCE	70 000	6 000

department	total-salary	total-bonus
IT	122 000	10 500
HR	109 000	7 500
FINANCE	70 000	6 000

12.

```
SELECT department,
        avg - salary
FROM Salaries
GROUP BY department
HAVING avg - salary > 55 000;
```

department	avg - salary
IT	61 000
FINANCE	70 000

13.

```
SELECT employee - id,
        Name,
        Salary,
        bonus,
        total - compensation
FROM Salaries
WHERE (Salary + bonus) > 60 000;
```

employee - id	Name	Salary	bonus	total - compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

SECTION 5

14.

```
SELECT department,
        total-budget,
        avg-budget
FROM    projects
GROUP BY department
HAVING avg-budget > 70 000;
```

department	total-budget	avg-budget
IT	270 000	135 000
FINANCE	80 000	80 000

15.

```
SELECT project-id,
        project-name,
        department,
        budget
FROM    projects
WHERE   budget BETWEEN 50 000 AND 120 000 AND
        department != 'Marketing';
```

project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	FINANCE	80 000
5	HR Portal	HR	50 000